Zia Farooq PhD researcher / Data scientist

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Education

PhD in Epidemiology and Public Health
Umeå University
Feb 2020 – Jun 2024 | Umeå, Sweden
Infectious disease modelling, data science

Researcher
Institute of Fundamental Technological Research, Polish Academy of Sciences
Aug 2018 – Dec 2019 | Warsaw, Poland
Systems Biology

Masters in Applied Mathematics (18-years)
Institute of Space Technology Islamabad
2015 – 2017 | Islamabad, Pakistan
Nonlinear dynamical systems and bifurcation theory

Masters in Mathematics (16-years)
University of Gujrat
2007 – 2009 | Gujrat, Pakistan
Pure mathematics

Bachelors in Mathematics and Computer science
Punjab University
2004 – 2006 | Lahore, Pakistan

Research Experience/Skills

Data science | Modelling | Epidemiology | Biostatistics
Umeå, Sweden

- Working as a climate and health data scientist
- Big data wrangling, processing, merging, analysis, and visualization
- Applying advanced mathematical modelling techniques to simulate, assess and predict the risk of infectious diseases
- Developing methods to improve the disease severity metrics for novel infectious diseases
- Using machine learning and explainable AI to disentangle the drivers of emerging infectious diseases.
- Using biostatistics and epidemiological methods in disease modelling
- Climate-related disease expansion risk projections by using novel machine learning techniques and mechanistic models
- Developing European-level indicators for West Nile virus (WNV) disease for Lancet Countdown-Europe
- Writing and publishing the research results
- Reviewer-Environmental Health journal
- Grant and research proposals writing

Publications

The 2024 Europe report of the Lancet Countdown on health and climate change: unprecedented warming demands unprecedented action
Lancet Public Health,
12 May 2024
Kim R van Daalen, Cathryn Tonne, Jan Semenza, ..., Zia Farooq, ... Rachel Lowe

Decision-support tools to build climate resilience against emerging infectious diseases in Europe and beyond
Lancet Regional Health-Europe, VOLUME 32, 100701
Sep 2023

European projections of West Nile virus transmission under climate change scenarios
One Health, Volume 16, 100509
Feb 2023
Zia Farooq, Henrik Sjödin, Jan C Semenza, Jonas Wallin, Yesim Tozan, Maquines Sewe, Joacim Rocklöv

The 2022 Europe Report of the Lancet Countdown on Health and Climate Change: towards a more climate resilient future
Lancet Public Health, Volume 7, E942-E965
Oct 2022
Kim Robin van Daalen, Marina Romanello, Joacim Rocklöv, ..., Zia Farooq, ..., Rachel Lowe

Artificial intelligence to predict West Nile virus outbreaks with eco-climatic drivers
Lancet Regional Health-Europe, Volume 17, 100370
Mar 2022
Zia Farooq, Joacim Rocklöv, Jonas Wallin, Najmeh Abiri, Maquines Sewe, Henrik Sjödin, Jan C Semenza

COVID-19 healthcare demand and mortality in Sweden in response to non-pharmaceutical mitigation and suppression scenarios
International Journal of Epidemiology, Volume 49, 1443–1453
Sep 2020
Henrik Sjödin, Andres F Johansson, Åke Bränström, Zia Farooq, ..., Joacim Rocklöv

Information processing in unregulated and autoregulated gene expression
European control conference (ECC), Saint Petersburg, Russia
May 2020
Zahra Vahdat, Karol Nienałtowski, Zia Farooq, Michał Komorowski, Abhyudai Singh

Courses

Mosquito-borne pathogens and their vectors
2022 | Umeå University, Sweden

Regression models in medical science
2021 | Umeå University, Umeå, Sweden

Research methodology with biostatistics
2021 | Umeå University, Umeå, Sweden

Writing Science
2021 | Umeå University, Sweden

Bayesian Statistics
2019 | University of Warsaw, Warsaw, Poland

Programming, Numerics and Optimization
2019 | IFTR, Polish Academy of Sciences, Warsaw, Poland

Python programming for biologists
2019 | NEB, Polish Academy of Sciences, Warsaw, Poland

Nonlinear Dynamics and Chaos
2015 – 2016 | Institute of Space Technology, Islamabad, Pakistan

Training / Teaching

Course teaching: Modelling and Simulations
Masters Level
2022 | Umeå University, Sweden

DevRes2021 Conference Umeå University
Technical Intern
Jun 2021 | Umeå University, Sweden

Member steering committee
Young researchers network, UTRI
Dec 2020 – Dec 2021 | Umeå University, Sweden

Technical Skills

Programming
R, MATLAB, C++, SAS, SPSS, Python, shell scripting

Machine Learning / AI
Classification, Regression, Clustering, PCA, eXplainable AI, Boosting and Bagging, Survival Analysis, Hypothesis testing, Neural networks

Collaboration / virtual communication
Github, Google Docs, Zoom, Teams,

Only strict quarantine measures can curb the coronavirus disease (COVID-19) outbreak in Italy, 2020
Eurosurveillance, Volume 25, Issue 13
Apr 2020
Henrik Sjödin, Annelies Wilder-Smith, Sarah Osman, Zia Farooq, Joacim Rocklöv

Bifurcation analysis of heartbeat model
Journal of Space Technology
2017
Zia Farooq, Salman Ahmed

Improving estimates of case fatality ratios early in pandemics by accounting for case-to-death times distribution analysis
Under review
2024
Zia Farooq, Henrik Sjödin, Joacim Rocklöv, Åke Brännström

Conferences/Seminars

Swedish Medical Association conference
Poster presentation: Case fatality ratio
Dec 2020 | Digital, Sweden

2nd Nordic Biomathematics conference
Poster presentation
May 2022 | Roskilde University, Denmark

Seminar on "Mathematical modelling and its applications"
Keynote Speaker
Jun 2022 | University of Gujrat | Digital, Pakistan

Seminar on "One Health PACT modelling"
Invited talk
Jul 2022 | Leiden University, Netherlands

Polish Scientific Network (PSN) conference
Oral presentation: Climate change and pandemic risk
Sep 2022 | Wroclaw University, Poland

European Scientific Conference on Applied Infectious Disease Epidemiology (ESCAIDE), 2022
Participant
Nov 2022 | Stockholm, Sweden

Departmental Journal Club Seminar Series
Oral presentations
Monthly Recurring event |
Department of Public Health and Clinical Medicine, Umeå University, Sweden

Conference on "Urban spaces and mosquitoes on a changing planet"
Oral presentation: European projections of West Nile virus under climate change scenarios
Mar 2023 | Barcelona, Spain